

The "Digital Doorman" (Face Verification) using Machine Learning

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Problem & Data

Problem

The goal of this project is to verify whether two face images belong to the same person.

This is a 1:1 face verification problem, for example comparing an ID photo with a selfie.

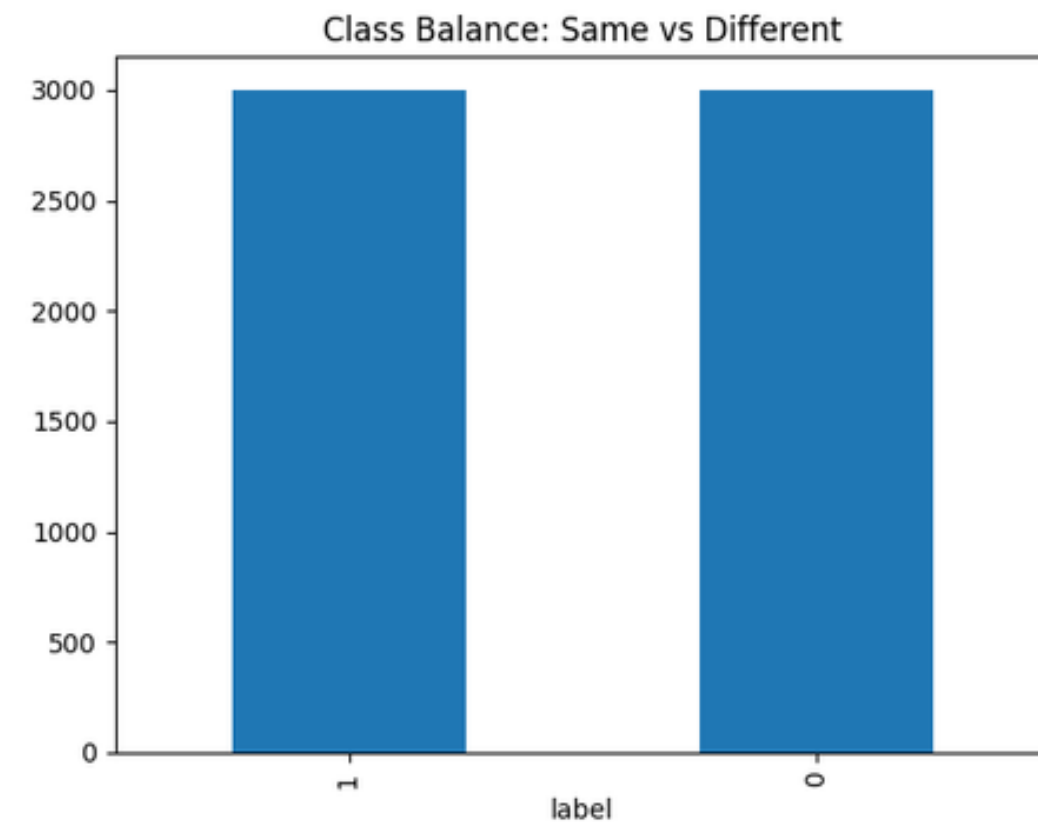
Data

LFW deep-funneled face dataset. Contains pairs of face images with labels: same person or different people.

Same person pairs: two images of the same identity

Different people pairs: images from two different identities

Exploratory Data Analysis

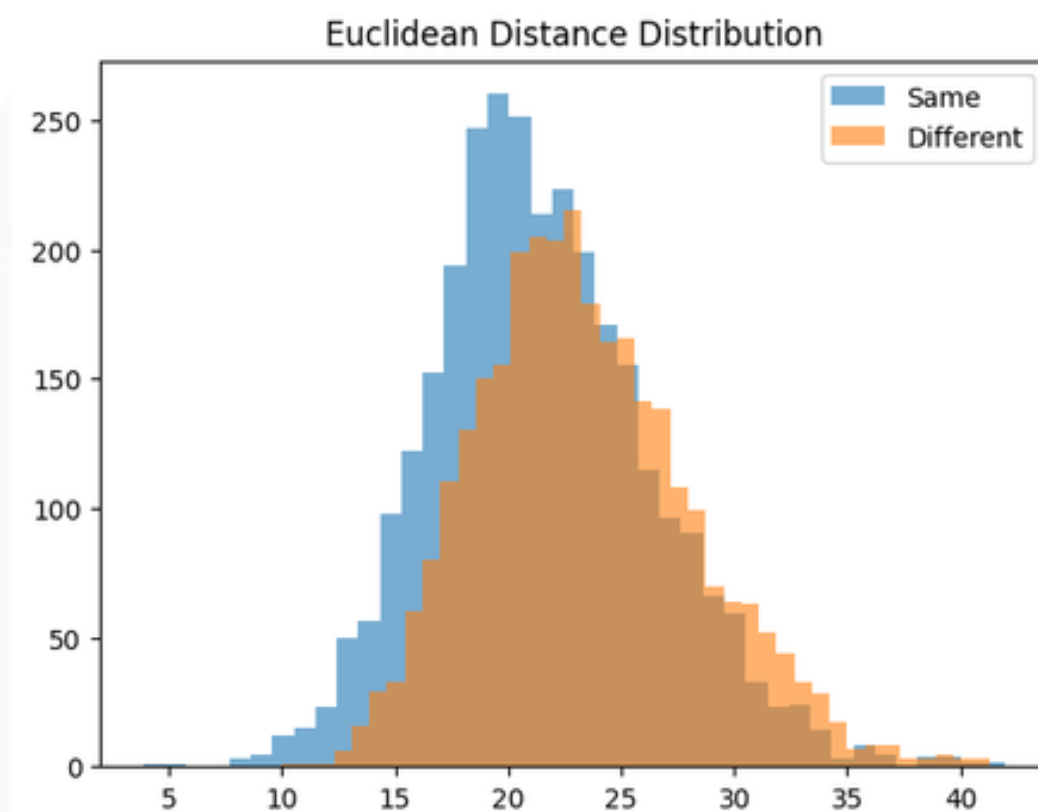


This chart shows the number of image pairs labeled as:

- same person
- different people

We can see that both classes have almost the same number of samples.

This is important because the model is not biased toward one class. It means that accuracy and F1-score are reliable and not misleading.



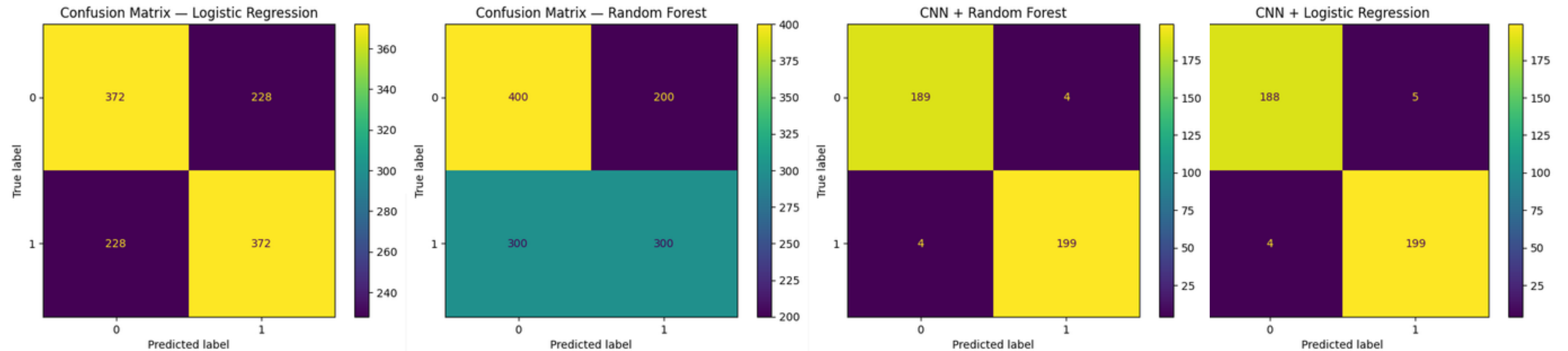
This chart shows the distribution of Euclidean distance for two classes:

- pairs of the same person
- pairs of different people

Distances for the same person are generally lower, while distances for different people are higher. There is some overlap between the two distributions, but overall the difference is visible.

This means that Euclidean distance is a useful feature

Model Comparison



This is a performance comparison of different models using confusion matrices.

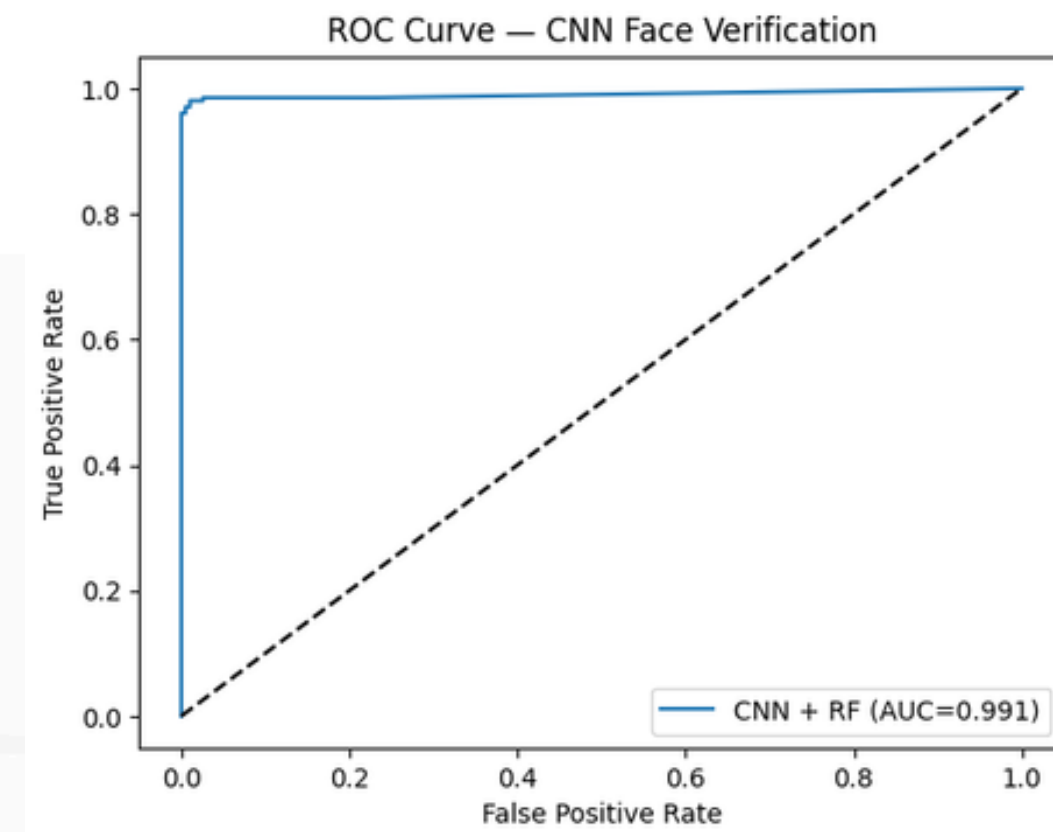
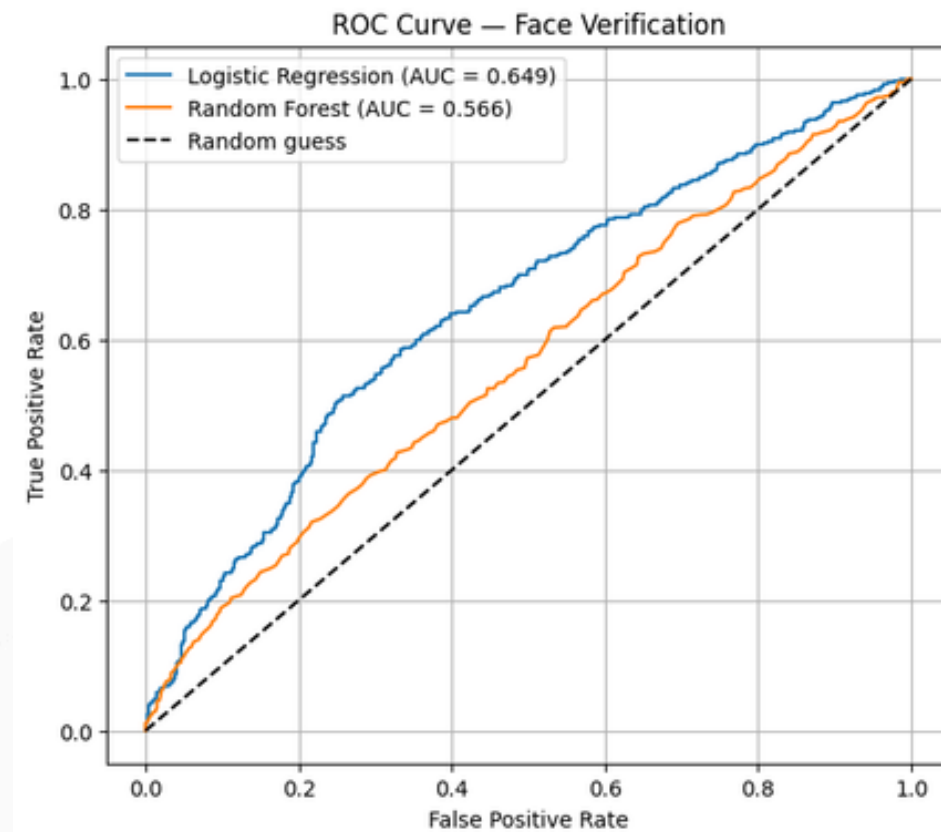
Logistic Regression and Random Forest use simple features and show a noticeable number of misclassifications.

After using CNN embeddings, the number of errors is significantly reduced.

Both CNN-based models correctly classify most of the image pairs.

Overall, CNN + Random Forest shows the best and most stable results.

Model Comparison



This compares model performance using ROC curves.

The Logistic Regression with $AUC = 0.649$ and Random Forest with $AUC = 0.566$ shows limited performance. After using CNN embeddings, the model performance improves significantly.

The ROC curve for the CNN-based model is much closer to the top-left corner. The CNN + Random Forest model achieves an AUC close to 1.0, which means it can clearly separate the two classes.

Prediction

Prediction: SAME person

Probability: 1.00

Similarity features:

Cosine similarity: 0.934

Euclidean distance: 0.525